

VIBROACOUSTICS - ASSIGNMENT 2

Bending vibrations of thin plates

Francesco Bandera, Anna Impembo
Master in Music and Acoustic Engineering
Politecnico di Milano

July 2025

Contents

1	Simply Supported Plate Natural Frequencies and Mode Shapes	2
1.1	Theoretical Derivation	2
1.2	Natural Frequencies Computation	3
1.3	Mode Shape (5,4) Computation and Plot	3
2	Modal Approach for Plate Dynamics	4
2.1	Motivation and General Strategy	4
2.2	Mathematical Formulation	4
2.3	Computation of Modal Matrices	4
2.4	Frequency Response at Arbitrary Points	5
3	Point and Transfer Mobility	6
3.1	Theoretical Definition of Mobility	6
3.2	Point Mobility at Point D	6
3.3	Transfer Mobility at Point C	7
4	Vibration Velocity Field at 440 Hz	8
4.1	Objective and Approach	8
4.2	Mathematical Formulation	8
4.3	Implementation and Visualization in MATLAB	8

1 Simply Supported Plate Natural Frequencies and Mode Shapes

1.1 Theoretical Derivation

We consider the bending vibrations of a thin rectangular plate, simply supported on all edges, described by Kirchhoff-Love theory. The transverse displacement $w(x, y, t)$ is assumed to follow a standing wave solution of the form:

$$w(x, y, t) = \Phi(x, y)G(t) \quad (1)$$

This form represents synchronous motion in which the spatial profile $\Phi(x, y)$ does not change in time, and $G(t)$ carries the temporal oscillation.

Substituting this form into the plate equation:

$$\nabla^4 w(x, y, t) = -\frac{\rho s}{B} \frac{\partial^2 w}{\partial t^2} \Rightarrow \nabla^4 \Phi(x, y) + \frac{\rho s \omega^2}{B} \Phi(x, y) = 0 \quad (2)$$

assuming $G(t) = G_0 e^{j\omega t}$.

We now separate variables in the spatial mode shape function:

$$\Phi(x, y) = X(x)Y(y) \quad (3)$$

Applying the first four simply supported boundary conditions on the edges parallel to the y -axis (i.e., $x = 0$ and $x = a$):

$$\Phi(0, y) = 0, \quad \frac{\partial^2 \Phi}{\partial x^2}(0, y) = 0, \quad \Phi(a, y) = 0, \quad \frac{\partial^2 \Phi}{\partial x^2}(a, y) = 0 \quad (4)$$

leads to:

$$X(x) = \sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, 3, \dots \quad (5)$$

Now, treating $Y(y)$ as unknown, and substituting $\Phi(x, y) = \sin\left(\frac{n\pi x}{a}\right) F(y)$ into the plate equation, we derive an ODE for $F(y)$. Solving the resulting characteristic equation leads to a general solution for $F(y)$ as a linear combination of trigonometric and hyperbolic functions:

$$F(y) = A \sin(\gamma y) + B \cos(\gamma y) + C \sinh(\gamma y) + D \cosh(\gamma y) \quad (6)$$

Applying the remaining two simply supported conditions on the edges $y = 0$ and $y = b$ eliminates constants B and D , and imposes the condition that $\sin(\gamma b) \sinh(\gamma b) = 0$, from which we deduce:

$$\gamma = \frac{m\pi}{b}, \quad m = 1, 2, 3, \dots \quad (7)$$

The final expression for the spatial mode shapes is:

$$\Phi_{n,m}(x, y) = \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \quad (8)$$

The associated natural frequencies are given by:

$$\omega_{n,m} = \left[\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2 \right] \sqrt{\frac{B}{\rho s}} \quad (9)$$

where:

$$B = \frac{Es^3}{12(1-\nu^2)}$$

1.2 Natural Frequencies Computation

Using the formula above, the first 100 natural frequencies (modes $n = 1 \dots 10$, $m = 1 \dots 10$) were computed in MATLAB. Below is the table of results:

freq_nat =									
1.0e+03 *									
0.0247	0.0840	0.1829	0.3214	0.4993	0.7169	0.9739	1.2706	1.6068	1.9825
0.0396	0.0989	0.1978	0.3362	0.5142	0.7317	0.9888	1.2854	1.6216	1.9973
0.0643	0.1236	0.2225	0.3609	0.5389	0.7564	1.0135	1.3101	1.6463	2.0220
0.0989	0.1582	0.2571	0.3955	0.5735	0.7910	1.0481	1.3447	1.6809	2.0566
0.1434	0.2027	0.3016	0.4400	0.6180	0.8355	1.0926	1.3892	1.7254	2.1011
0.1978	0.2571	0.3560	0.4944	0.6724	0.8899	1.1470	1.4436	1.7798	2.1555
0.2620	0.3214	0.4202	0.5587	0.7366	0.9542	1.2112	1.5079	1.8441	2.2198
0.3362	0.3955	0.4944	0.6328	0.8108	1.0283	1.2854	1.5820	1.9182	2.2940
0.4202	0.4796	0.5784	0.7169	0.8948	1.1124	1.3695	1.6661	2.0023	2.3780
0.5142	0.5735	0.6724	0.8108	0.9888	1.2063	1.4634	1.7600	2.0962	2.4719

Figure 1: Table of natural frequencies $f_{n,m} = \omega_{n,m}/2\pi$ [Hz] for a simply supported plate

1.3 Mode Shape (5,4) Computation and Plot

Using the previously derived formula yields:

$$\Phi_{5,4}(x, y) = \sin\left(\frac{5\pi x}{a}\right) \sin\left(\frac{4\pi y}{b}\right) \quad (10)$$

This function was evaluated on a 2D spatial grid and visualized both using a 2D color plot and a 3D surface plot, generated in MATLAB. The results are shown in the figure below.

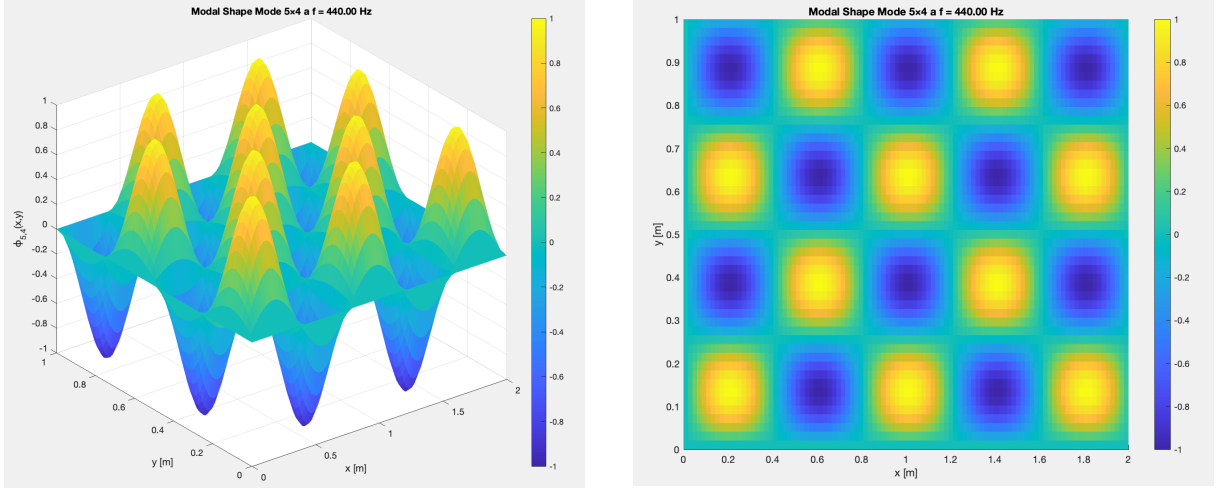


Figure 2: Mode shape $\Phi_{5,4}(x, y)$ visualized using two complementary 3D surface plots

The mode (5,4) corresponds to a natural frequency of approximately 440 Hz, which matches the excitation frequency later used in the forced vibration analysis. In modal analysis, when an external harmonic force matches one of the system's natural frequencies, the associated mode becomes dominant in the response. Therefore, mode (5,4) is expected to contribute the most to the steady-state vibration at 440 Hz, making it a physically meaningful and representative choice for visualization. Nodal lines (where the displacement is always zero) form a regular grid, dividing the plate into regions of positive and negative amplitude. This structure reflects the orthogonality and regularity of the eigenmodes for a simply supported rectangular plate.

2 Modal Approach for Plate Dynamics

2.1 Motivation and General Strategy

The modal approach is a well-established method for solving the vibration response of continuous systems like plates. It enables the transformation of the original partial differential equation into a system of decoupled ordinary differential equations in the modal domain. This is particularly effective when the structure is excited at an internal point (not on a boundary), as is the case where the reaction force $R(\omega)$ from the string is applied at point $D = (0.9, 0.6)$.

The general time-domain displacement is expressed as:

$$w(x, y, t) = \sum_{i=1}^N \phi_i(x, y) \cdot q_i(t) \quad (11)$$

where $\phi_i(x, y)$ are the spatial mode shapes and $q_i(t)$ are the modal coordinates.

Assuming a steady-state harmonic excitation, the modal coordinates are of the form:

$$q_i(t) = \hat{q}_i e^{j\omega t} \quad (12)$$

and the displacement becomes:

$$w(x, y, t) = \sum_{i=1}^N \phi_i(x, y) \cdot \hat{q}_i e^{j\omega t} \quad (13)$$

2.2 Mathematical Formulation

For each mode, the modal equation in presence of damping and forcing becomes:

$$(-\omega^2 m_i + j\omega c_i + k_i) q_i(\omega) = Q_i(\omega) \quad (14)$$

where m_i , c_i , and k_i are the modal mass, damping, and stiffness, and Q_i is the modal force:

$$Q_i(\omega) = \phi_i(x_D, y_D) \cdot R(\omega) \quad (15)$$

2.3 Computation of Modal Matrices

We define the mass per unit area as:

$$m = \rho \cdot s = 3.279 \text{ kg/m}^2 \quad (16)$$

The mass matrix $[\mathbf{M}]$ is diagonal due to the orthogonality of the modes and can be expressed as:

$$[\mathbf{M}] = \begin{bmatrix} \iint m \phi_1 \phi_1 \, dA & \iint m \phi_2 \phi_1 \, dA & \cdots & \iint m \phi_{100} \phi_1 \, dA \\ \iint m \phi_1 \phi_2 \, dA & \iint m \phi_2 \phi_2 \, dA & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \iint m \phi_1 \phi_{100} \, dA & \cdots & \cdots & \iint m \phi_{100} \phi_{100} \, dA \end{bmatrix} \quad (17)$$

Using the relation $k_i = m_i \omega_i^2$, the stiffness matrix $[\mathbf{K}]$ is:

$$[\mathbf{K}] = \begin{bmatrix} m_1 \omega_1^2 & 0 & \cdots & 0 \\ 0 & m_2 \omega_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_{100} \omega_{100}^2 \end{bmatrix} \quad (18)$$

The damping matrix $[\mathbf{C}]$ is calculated using Rayleigh damping as a linear combination of the mass and stiffness matrices:

$$[\mathbf{C}] = \alpha[\mathbf{M}] + \beta[\mathbf{K}] \quad \text{with } \alpha = 0.01, \beta = 1 \times 10^{-5} \quad (19)$$

Thus, $[\mathbf{C}]$ becomes:

$$[\mathbf{C}] = \begin{bmatrix} m_1(\alpha + \beta \omega_1^2) & 0 & \cdots & 0 \\ 0 & m_2(\alpha + \beta \omega_2^2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_{100}(\alpha + \beta \omega_{100}^2) \end{bmatrix} \quad (20)$$

All quantities were evaluated numerically in MATLAB using $N = 100$ modes.

2.4 Frequency Response at Arbitrary Points

Once the modal parameters are known, the frequency response (velocity per unit force) at any point (x, y) on the plate due to a force $R(\omega)$ at point D is given by:

$$v(x, y,) = j\omega \cdot Re^{j\omega t} \cdot \sum_{i=1}^N \frac{\phi_i(x, y) \cdot \phi_i(x_D, y_D)}{-\omega^2 m_i + j\omega c_i + k_i} \quad (21)$$

This formula was used to compute the point mobility at D and the transfer mobility at point $C = (0.6, 0.6)$, as detailed in the next section.

3 Point and Transfer Mobility

3.1 Theoretical Definition of Mobility

Accordingly to the modal approach, we can define the vibration velocity of a generic point on the plate $(\tilde{x}, \tilde{y})$ as the superposition of the first 100 modes with respect to the excitation point D . In order to define the *point mobility* and the *transfer mobility*, we evaluate the velocity at the point $C = (0.6, 0.6)$ (sensor location) and at the point $D = (0.9, 0.6)$ (drive attachment):

$$v_D(x_D, y_D) = j\omega \cdot Re^{j\omega t} \cdot \sum_{i=1}^{100} \frac{\phi_i(x_D, y_D)^2}{-\omega^2 m_i + j\omega c_i + k_i} \quad (22)$$

$$v_C(x_C, y_C) = j\omega \cdot Re^{j\omega t} \cdot \sum_{i=1}^{100} \frac{\phi_i(x_D, y_D)\phi_i(x_C, y_C)}{-\omega^2 m_i + j\omega c_i + k_i} \quad (23)$$

where R is the reaction force previously calculated.

The corresponding mobility functions with respect to the input force F are then obtained by dividing the velocities by F :

$$\text{Point Mobility} = \frac{v_D(x_D, y_D)}{F} \quad (24)$$

$$\text{Transfer Mobility} = \frac{v_C(x_C, y_C)}{F} \quad (25)$$

3.2 Point Mobility at Point D

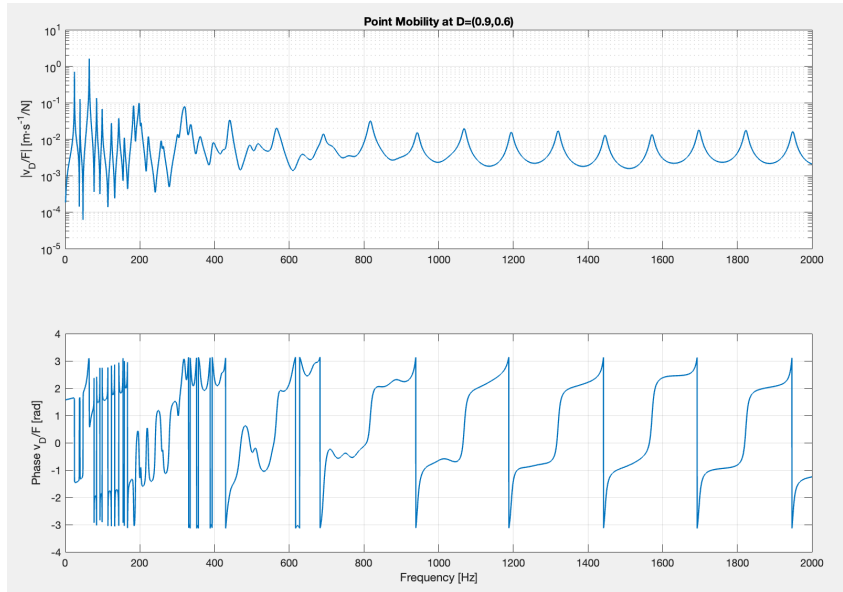


Figure 3: Point mobility $|v_D/F|$ and phase at $D = (0.9, 0.6)$

The point mobility plot shows the frequency response of the plate at the excitation point D.

The magnitude spectrum reveals a series of peaks at the plate's natural frequencies, corresponding to resonances where the velocity response is maximized for a unit force input. At low frequencies, the modal contributions tend to overlap more significantly, resulting in a denser and less regular spectral structure. As frequency increases, the modal spacing becomes more uniform and the resonance peaks sharper and more separated, clearly highlighting the plate's high-order modal characteristics.

The phase plot shows sharp transitions around the resonances, shifting by approximately π radians. This behavior is typical of linear second-order systems, reflecting the transition from in-phase to out-of-phase response relative to the excitation.

3.3 Transfer Mobility at Point C

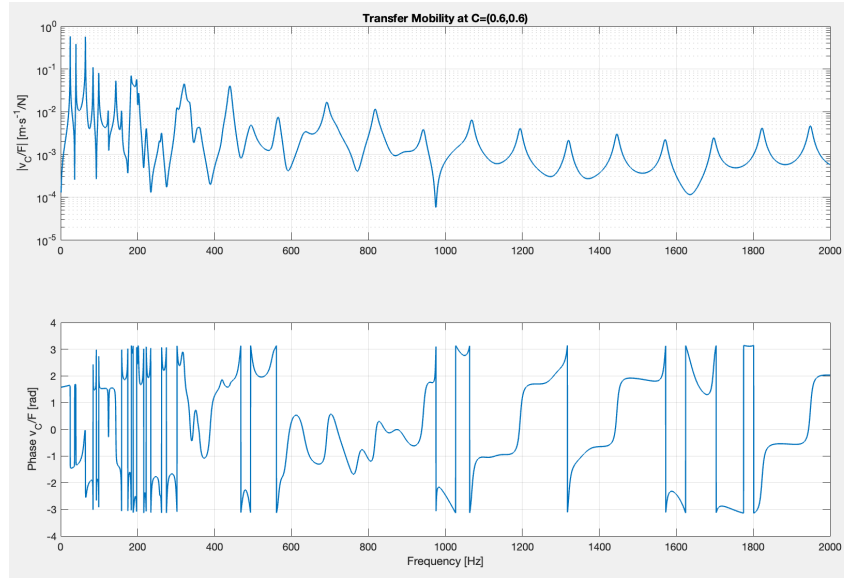


Figure 4: Transfer mobility $|v_C/F|$ and phase from D to $C = (0.6, 0.6)$

The transfer mobility plot displays the response measured at point C due to excitation at point D. The overall structure is similar to the point mobility, with clear resonance peaks at the same frequencies, but the amplitudes are generally lower. This is because the transfer mobility depends on the spatial correlation between the mode shapes at the two different points, and not all modes contribute equally to the response at C. In particular, the modal contribution is weighted by the product $\Phi_i(C)\Phi_i(D)$, which can be small or even vanish if either point lies close to a nodal line of mode i .

The phase behavior is qualitatively similar to the point mobility, with sharp π jumps at resonance frequencies. However, the presence of both positive and negative contributions in the modal summation introduces more complex phase variations, especially at low frequencies where modal overlap is more pronounced.

Both point and transfer mobility functions highlight the resonance structure of the plate, but the point mobility at the drive location exhibits higher amplitudes and more regular behavior, as each mode is optimally excited and measured at its antinode. The transfer mobility, by contrast, reflects the spatial filtering effect of the measurement point, with some resonances diminished or even suppressed if a nodal line passes through point C. At higher frequencies, both mobility functions become more regular as the modes become more spatially dense and the effect of individual nodal lines at specific points is reduced. These resonance features in the plate's mobility are directly related to the behavior observed in the reaction force spectrum of the string in the previous lab (Figure 5). In particular, the sharp peaks in the normalized reaction force amplitude indicate frequencies where the force transmitted from the string to the plate is maximized—these are precisely the frequencies at which the plate is most strongly excited and its mobility is most pronounced. This demonstrates how the dynamic response of the string, through the reaction force at the coupling point, governs the excitation of the plate and thus shapes the overall vibroacoustic behavior of the coupled system.

4 Vibration Velocity Field at 440 Hz

4.1 Objective and Approach

The aim of this section is to compute and visualize the vibration velocity field $v(x, y)$ on the surface of the simply supported plate when the string is excited by means of a unitary force with frequency of 440 Hz. This frequency corresponds closely to the natural frequency of the (5, 4) mode, which is expected to dominate the response.

To this end, we employ the modal superposition method. The total velocity field is obtained by summing the contribution of the first 100 modes weighted by the modal response under harmonic excitation at f_0 .

4.2 Mathematical Formulation

From the modal approach, the velocity at any point (x, y) on the plate is given by:

$$v(x, y, \omega_0) = j\omega_0 \cdot R(\omega_0) \cdot \sum_{i=1}^N \frac{\phi_i(x, y) \cdot \phi_i(x_D, y_D)}{-\omega_0^2 m_i + j\omega_0 c_i + k_i} \quad (26)$$

where:

- $\omega_0 = 2\pi f_0$ is the angular frequency corresponding to 440 Hz
- $R(\omega_0)$ is the complex-valued reaction force calculated in Lab01

The double product $\phi_i(x, y) \cdot \phi_i(x_D, y_D)$ ensures that only the modes active at the excitation point and at the observation point contribute effectively.

4.3 Implementation and Visualization in MATLAB

The velocity field $v(x, y)$ was evaluated over a grid of 100×50 points spanning the plate domain $[0, a] \times [0, b]$ using the following MATLAB procedure:

1. Interpolate $R(\omega)$ from Lab01 to extract $R(\omega_0)$
2. Loop over all $N = 100$ modes to compute the modal response at $f_0 = 440$ Hz
3. Sum each mode's contribution scaled by its spatial modal shape over the grid
4. Multiply the result by $j\omega_0 R(\omega_0)$ to obtain the velocity field

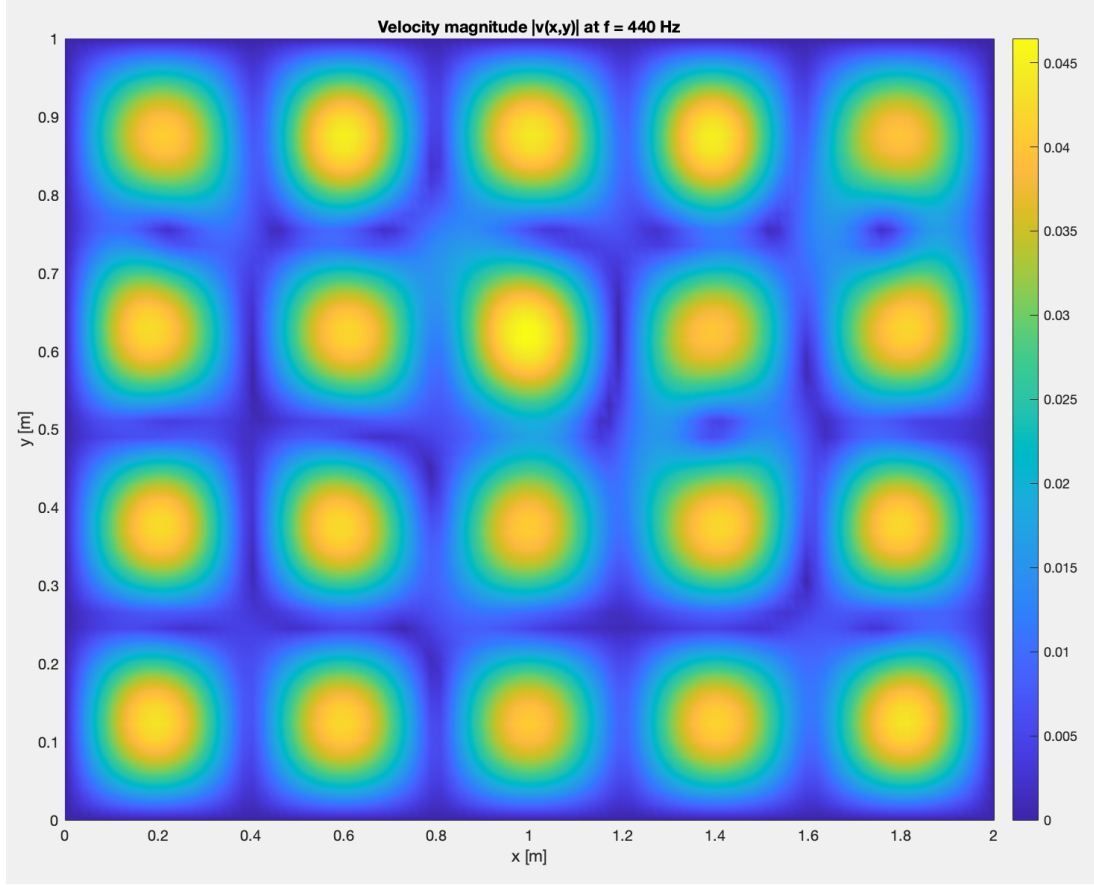


Figure 5: Magnitude of the vibration velocity field $|v(x, y)|$ at 440 Hz

The resulting plot clearly reveals the spatial structure of the steady-state velocity field induced by the excitation at $f = 440$ Hz. Regions of maximum velocity (in yellow tones) correspond to antinodal areas, where the plate exhibits the strongest vibratory response. Conversely, dark blue regions represent nodal lines—zones of minimal or zero motion—resulting from destructive modal interference.

The pattern closely matches the modal shape of the $(n = 5, m = 4)$ mode, which is strongly excited due to its natural frequency being near 440 Hz. However, the response is not purely limited to a single mode: minor contributions from neighboring modes are visible as subtle asymmetries and irregularities in the field. This illustrates the physical principle that, under harmonic excitation, all modes contribute according to their spatial projection and dynamic amplification factor.

Additionally, we observe that the location of highest vibration velocity coincides with an antinode situated near the excitation point $D = (0.9, 0.6)$, confirming the localized energy input and its spatial effect on the modal superposition. This also reflects the spatial filtering property of the excitation point—only those modes with significant amplitude at D can be effectively excited.

Overall, the field confirms both the theoretical modal structure of the plate and the frequency-selective nature of its dynamic response.